

F971 — CORRECTION of F970, on Casey’s catch: my “two right-handed neutrinos” reads as a *physical* claim, and taken literally it contradicts the Five-Absence set (“NO sterile neutrinos”). Casey is right. The resolution is his option 2: the RH neutrinos are mathematical bookkeeping only — the physical mechanism is a geometric, rank-2 Majorana mass matrix with no sterile states, and Five-Absence stands untouched. ★ The tension (Casey caught it). The Five-Absence engine is D_{IV}^5 irreducibility — the domain has no room for extra fermion states (hence no DM particle, no sterile ν , no SUSY partners). A physical right-handed neutrino is a sterile state (an SM-singlet fermion), so the same irreducibility that forbids sterile ν forbids physical RH neutrinos. F970 wrote “the number of heavy right-handed neutrinos = rank = 2” as if two real states exist — that contradicts the Five-Absence “no sterile neutrinos.” Owned. ★ The resolution — Casey’s option 2 (mathematical only). The observable ($m_{\nu 1} = 0$ exactly) requires *only* that the light-neutrino Majorana mass matrix has rank 2 (a one-dimensional kernel). That rank-deficiency is a property of the geometric mass matrix itself — it does not require integrating out physical sterile states. Verified: a rank-2 symmetric 3×3 Majorana mass, built with no RH neutrino (just a rank-deficient geometric matrix = sum of two rank-1 pieces), already gives one exactly-massless neutrino (kernel eigenvalue $\sim 10^{-19}$, machine-zero) plus two massive ones. Rank 2 is a matrix fact, not a particle count. So the “seesaw with 2 RH neutrinos” (F970 Route B) was the *standard bookkeeping picture* for a rank-2 matrix — I retire it as a physical claim. ★ The physical mechanism (Five-Absence-consistent). The neutrino mass is geometric — the Majorana mass is generated directly by the D_{IV}^5 structure (the Weinberg operator from the geometry / the Shilov-boundary Majorana structure), with no new states. That geometric mass matrix has rank 2 (= the BST rank), giving a one-dimensional kernel = one exactly-massless neutrino, and the kernel is protected by the Z_3 generation symmetry (Keeper’s Route A — now the *sole* physical route). No sterile neutrinos, no extra fermions: Five-Absence stands. ★ What survives unchanged. The *observables* are untouched — they ride the rank-2 kernel, which survives the reinterpretation: $m_{\nu 1} = 0$ exactly, normal ordering, $\Sigma m_{\nu} \approx 0.059$ eV floor, sub-4-meV $0\nu\beta\beta$ (the kill-test). Only the *mechanism-picture* is corrected: geometric rank-2 mass (no sterile states), not a physical two-RH seesaw. And it still fits the session’s counting engine — “exactly massless = the one-dimensional kernel of a rank-2 count” — but the count is a property of a geometric matrix, not a tally of particles. ★ Honest note. This is the good kind of catch: Casey held the new mechanism against a banked principle (Five-Absence), found the collision, and it forced the *better* (geometric, state-free) formulation. The rank-2-from-geometry version is not a patch — it’s cleaner: the massless neutrino is a kernel of the ge-

The correction, in one line

F970: “2 physical RH neutrinos” (contradicts Five-Absence) → F971: geometric rank-2 Majorana mass, no sterile states, Z_3 -protected kernel (Five-Absence stands). Observable ($m_{\nu 1} = 0$, $0\nu\beta\beta$ window) unchanged.

RUBRICS Layer-2 done-bar (correction, honest)

- ☒ Owned Casey’s catch: F970’s “2 RH neutrinos” reads as physical, contradicting Five-Absence (“no sterile ν ”); D_{IV}^5 irreducibility forbids physical RH/sterile states.
- ☒ Resolved via option 2: RH neutrinos are mathematical bookkeeping; the physical content is a geometric rank-2 Majorana mass (verified: rank-2 matrix gives one massless with NO RH neutrino).
- ☒ Restated the mechanism Five-Absence-consistently: geometric Majorana mass (no new states), rank 2 = BST rank, one-dim kernel = massless, Z_3 -protected (Route A sole physical route).
- ☒ Confirmed the observables survive ($m_{\nu 1}=0$, normal ordering, Σm_{ν} floor, sub-4-meV $0\nu\beta\beta$) — ride the kernel, unchanged.
- ☒ Retired F970 Route B (2-RH seesaw) as a physical claim; kept it only as bookkeeping. Nothing pushed.

Handoffs

- @Keeper — F970 corrected per Casey’s Five-Absence catch. “Two RH neutrinos” was a bookkeeping picture read as physical → contradicts “no sterile ν .” Retired as physical. The mechanism is now: geometric rank-2 Majorana mass (no new states), Z_3 -protected kernel → $m_{\nu 1} = 0$ exactly. Your Z_3 -Goldstone route (Route A) is the *sole physical* route; the rank-2 seesaw (Route B) is bookkeeping only. Five-Absence stands; the observables ($0\nu\beta\beta$ window, Σm_{ν} floor) are unchanged. Please flag F970’s Route-B-as-physical superseded in the ledger (never silent-delete).
- @Grace — the mechanism behind your $0\nu\beta\beta$ kill-test, corrected and cleaner. The lightest neutrino is exactly massless because the geometric Majorana mass matrix has rank 2 (a one-dimensional kernel) — not because of physical sterile neutrinos (which Five-Absence forbids). The observable is identical (sub-4-meV window, from the rank-2 kernel); only the interpretation is now state-free/geometric. Your kill-test is untouched and arguably stronger (no added particles to hedge with).
- @Cal — a correction to bank. F970’s “RH-count = rank = 2” (physical) is retired; it contradicted Five-Absence. The corrected claim: the geometric neutrino Majorana mass has rank 2 (one massless, Z_3 -protected) with no sterile states. The load-bearing derivation is now “the geometry forces $\text{rank}(m_{\nu}) = 2$ ” (why the geometric mass matrix is rank-deficient by exactly one), which the Z_3 generation symmetry supplies — that’s the piece to certify when the mechanism firms up.
- @Casey — you were right, and the fix made it better, not worse. I did overstep: writing “two right-handed neutrinos” as if they’re real particles collides head-on with our own Five-Absence rule that says the geometry has no room for sterile states. Your second reading is the correct one — they’re mathematical, a bookkeeping trick for a mass matrix that’s “rank two.” Here’s the clean version: the neutrinos get their mass straight from the shape of the space, no new particles added; that mass pattern happens to be *rank two* — meaning it can give mass to only two of the three, and the third falls in the gap and stays exactly massless. No sterile neutrino anywhere; the “two” is just how many the geometry can lift, and it equals our rank. So the massless neutrino is a hole in the geometry’s own mass matrix, not the shadow of a hidden particle — which is exactly where you want our predictions to come from. The kill-test (the narrow double-beta window) is untouched, and honestly it’s *stronger* now, because there are no extra particles to hide behind. Thank you for holding the new idea against the old rule — that’s the check working. Nothing pushed.

Notes only; no toy/theorem claimed (correction of F970). F971: CASEY’S CATCH — F970 “2 physical RH neutrinos” CONTRADICTS Five-Absence “NO sterile neutrinos” (D_{IV}^5 irreducibility forbids extra fermion states; physical RH ν = sterile state). RESOLUTION = Casey option 2: RH neutrinos MATHEMATICAL ONLY (bookkeeping for rank-2 matrix). Physical mechanism: GEOMETRIC Majorana mass (Weinberg op from D_{IV}^5 / Shilov-boundary, NO new states) has RANK 2 (= BST rank) → one-dim KERNEL → $m_{\nu 1}=0$ exactly; kernel Z_3 -protected (Route A sole physical route). VERIFIED: rank-2 symmetric 3x3 Majorana (no RH neutrino, sum of 2 rank-1) → one massless ($\sim 1e-19$) + 2 massive. Rank 2 = MATRIX FACT not particle count. F970 Route B (2-RH seesaw) RETIRED as physical, kept as bookkeeping. Five-Absence STANDS. Observables UNCHANGED

($m_{\nu 1}=0$, normal ordering, Sum ~ 0.059 floor, sub-4-meV 0nubb) — ride the kernel. Massless = HOLE in geometry's mass matrix, not shadow of hidden particle. Keeper: flag F970 Route-B-physical superseded (never silent-delete); Z_3 = sole physical route. Grace: 0nubb mechanism = rank-2 geometric kernel, no sterile, observable identical/stronger. Cal: load-bearing = "geometry forces rank(m_{ν})=2" (Z_3 supplies), certify when firm. Nothing pushed. — Lyra